

Name: _____ Date: _____

Period: _____

1. Complete the difference table and identify the best model type.

x	y	Δy	$\Delta^2 y$
0	5	—	—
1	9	_____	—
2	17	_____	_____
3	29	_____	_____
4	45	_____	_____

The _____ differences are constant, so the best model type is _____ (degree _____).

2. A shipping company charges a flat rate of \$8 for packages up to 5 lb, then adds \$1.50 per pound above 5 lb. Circle the best model type and give one reason.

A. Linear B. Quadratic C. Cubic D. Piecewise-defined

One reason: _____

3. A quadratic function models the area of a rectangular garden with one fixed wall (using 60 m of fencing for three sides). State one **domain restriction** and one **model assumption**.

Domain restriction: _____

Model assumption: _____